Proposition: If Δ is a simplicial d-manifold, then for all nonempty $\sigma \in \Delta$

$$H_\bullet(\operatorname{lk}_\Delta \sigma) \cong H_\bullet(S^{d-1-\dim\sigma})$$

<u>Corollary</u>: For Δ a simplicial d-manifold and all nonempty $\sigma \in \Delta$.

$$\sum_{i \geq -1} (-1)^i f_i(\mathrm{lk}_\Delta \sigma) = (-1)^{d-1-\dim\sigma}$$

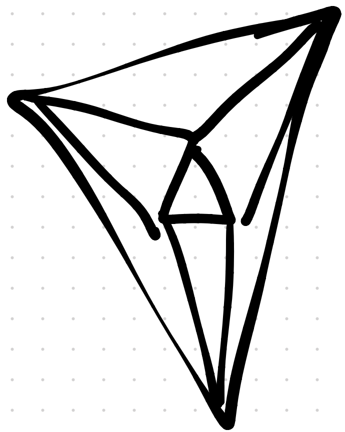
reduced Euler characteristic of $S^{d-1-\dim\sigma}$

$$\sum_{\tau \in \mathrm{lk}_\Delta \sigma} (-1)^{\dim \tau} =$$

$$\sum_{\Delta \ni \sigma' \supset \sigma} (-1)^{\dim\sigma' - \dim\sigma - 1} = (-1)^{d-1-\dim\sigma}$$

$$\sum_{\Delta \ni \sigma' \supset \sigma} (-1)^{\dim\sigma'} = (-1)^d$$

Recall our constraints on f-vectors of 2-spheres.



A vector (f_0, f_1, f_2) is the f-vector of a 2-sphere if and only if

$$f_0 - f_1 + f_2 = 2$$

$$3f_2 = 2f_1$$

$$f_0 \geq 4$$

Fix some $0 \le k \le d$.

$$\sum_{\sigma \in \Delta^{(k)}} \sum_{\Delta \ni \sigma' \supset \sigma} (-1)^{\dim \sigma'} = (-1)^d f_k$$

k-dim faces in Δ

$$\sum_{\sigma' \in \Delta} \sum_{\Delta^{(k)} \ni \sigma \subset \sigma'} (-1)^{\dim \sigma'} = (-1)^d f_k$$

$$\sum_{\sigma' \in \Delta} (-1)^{\dim \sigma'} \sum_{\Delta^{(k)} \ni \sigma \subset \sigma'} 1$$

$$\sum_{\sigma' \in \Delta} (-1)^{\dim \sigma'} \binom{\dim \sigma' + 1}{k+1} = (-1)^d f_k$$

$$\underbrace{\sum_{j \geq k+1}}_{\text{or } j \geq 0} (-1)^j \binom{j+1}{k+1} f_j = (-1)^d f_k$$

Dehn-Somerville equations:

If Δ is a simplicial d-manifold, then for all $0 \le k \le d-1$

$$\sum_{j=0}^{d} (-1)^j \binom{j+1}{k+1} f_j = (-1)^d f_k$$

and

$$\sum_{j=0}^{d} (-1)^j f_j = \chi(\Delta)$$

3-sphere

DS-equations:

$$f_0 - f_1 + f_2 - f_3 = 0$$

$k=1$ $$\binom{1}{1} f_0 - \binom{2}{1} f_1 + \binom{3}{1} f_2 - \binom{4}{1} f_3 = -f_0$$

$$2f_0 - 2f_1 + 3f_2 - 4f_3 = 0$$

$k=2$ $$\cancel{\binom{1}{2} f_0} - \cancel{\binom{2}{2} f_1} + \binom{3}{2} f_2 - \binom{4}{2} f_3 = \cancel{-f_1}$$

$$3f_2 - 6f_3 = 0$$

(already follows from previous two equations) $f_2 = 2f_3$

For a 3-sphere,

$$f_0 - f_1 + f_2 - f_3 = 0$$

$$f_2 = 2f_3$$

The Dehn-Somerville equations are not independent. The solution space over $\mathbb{R}$ is an affine space of dimension $\lceil d/2 \rceil$.

In fact, for any given manifold M, this is the smallest affine space containing all f-vectors of simplicial complexes $\Delta \approx M$. In other words, any linear or affine equation that is satisfied by all f-vectors of simplicial complexes $\approx M$ must be a combination of the DS-equations.

For any simplicial complex Δ of dimension d, let $h_0(\Delta), h_1(\Delta), \ldots, h_{d+1}(\Delta)$ be the integers such that

h-vector

$$\sum_{i=-1}^{d} f_i(\Delta)(x-1)^{d-i} = \sum_{i=0}^{d+1} h_i(\Delta)x^{d+1-i}$$

Alternatively,

$$h_k(\Delta) = \sum_{i=-1}^{k-1} (-1)^{k-1-i} \binom{d-i}{k-1-i} f_i(\Delta)$$

We always have $h_0 = 1$.

The Dehn-Sommerville equations are equivalent to

$$h_k = h_{d+1-k} \quad \text{for } 1 \leq k \leq \lfloor d/2 \rfloor$$

$$h_{d+1} = (-1)^d \tilde{\chi}(\Delta)$$